Fractional Fourier-Based Frequency-Spatial–Spectral Prototype Network for Agricultural Hyperspectral Image Open-Set Classification

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***Abstract*— At present, hyperspectral image classification (HSIC) technology has been warmly concerned in all walks of life, especially in agriculture. However, existing classification methods operate under the closed-set assumption, which deviates from the real world with open properties. At the same time, there are more serious phenomena of different crops with similar spectrums and the same crops with different spectrum in agricultural hyperspectral data, which is also a great challenge to existing methods. In this work, a fractional Fourier-based frequency-spatial–spectral prototype network (FrFSSPN) is proposed to address the challenges of open-set HSIC in agricultural scenarios. First, fractional Fourier transform (FrFT) is introduced into the network to combine the information in the frequency domain with the spatial–spectral information, so as to expand the difference between different classes on the premise of ensuring the similarity between classes. Then, the prototype learning strategy is introduced into the network to improve the feature recognition capability of the network through prototype loss. Finally, in order to break the stubbornly closed-set property of the closed-set classification (CSC) method, the open-set recognition module is proposed. The difference between the prototype vector and the feature vector is used to judge the unknown class. Experiments on three agricultural hyperspectral datasets show that this method can effectively identify unknown classes without sacrificing the classification accuracy of closed-set, and has satisfactory classification performance.**

***Index Terms*— Agricultural hyperspectral image classification (HSIC), fractional Fourier transform (FrFT), open-set classification (OSC), prototype learning.**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain tens to hundreds of continuous spectral bands [1], [2], [3], [4], [5], [6] and are widely used in vegetation structure analysis, crop classification pest detection, and yield estimation [7], [8], [9], [10]. As the foundation of the development and progress of human society, the effective analysis of agricultural data is of strategic significance. Due to the rich spatial–spectral information, HSIs can effectively reflect the differences between different crops [11], [12], [13]. Therefore, the pixel-level HSI classification (HSIC) methods are very suitable for crop planting structure analysis [14]. In recent years, many novel and effective methods have been proposed and applied to HSIC [15], [16], [17].

In the past period, machine learning theory has been widely discussed and used in HSIC. As the classical and effective methods in machine learning, support vector machine (SVM), random forest (RF), K-nearest neighbor (KNN) and other methods have been maturely applied in HSIC [18], [19], [20], [21]. In HSIC, the SVM aims to find a hyperplane that maximizes the spacing between different classes, thus separating pixels from different classes. The RF improves classification performance by combining multiple decision trees based on feature subsets or data subsets. The KNN determines the category by comparing the similarity between the samples to be divided and the training samples, and does not require complex model training. The above methods have their own

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advantages and can achieve considerable classification effects. However, in agricultural HSIs, due to the diversity of crops, satisfactory HSIC performance cannot be obtained by shallow features. When two crops are clearly different, shallow features can reflect a clear difference. However, when two crops are similar but not identical (such as broad-leaf soybean and narrow-leaf soybean), the shallow features do not reflect their differences in detail. On the contrary, relying on shallow feature classification often misclassifies them into one class. In other words, these classical methods are unable to extract deeper representative features [22], [23].

In order to carry out advanced feature discovery engineering, deep learning (DL) is introduced into HSIC under the background of rapid development of computer science and artificial intelligence theory [24], [25], [26]. The application of convolutional neural network (CNN) makes the development of the HSIC technology a qualitative leap [27], [28], [29], [30]. Hu et al. [27] applied deep CNNs in the field of HSIC, marking the initial application of CNN in the field of HSI processing. Hamida et al. [31] applied 3DCNN to HSIC and achieved good performance. In addition, a residual network has also been applied to HSIC. Zhong et al. [32] proposed spatial–spectral residual networks (SSRNs) and introduced batch normalization, which achieved good results.

However, due to the redundant spatial–spectral information in HSIs, giving equal attention to all information amplifies the negative effects of noncritical information and noise. For this, how to extract the key feature from the rich spatial–spectral information of HSIs has become a research focus. Inspired by visual attention, the attention mechanism has been applied to HSIC with good results [33], [34]. Li et al. [35] proposed a double-branch double-attention mechanism network (DBDA) to learn to extract spatial–spectral local key features of images by combining 3-D convolution with the dual attention mechanism network. Wu et al. [36] used cross-channel dense connection and multiscale double-attention (CDC-MDAA) to extract the deep features of HSIs.

However, the local perceptual properties of CNNs may limit their modeling of global relationships, which may lead to some information loss in HSIC. Therefore, in order to better capture the complex relationship between different bands in HSIs, transformer with global relationship modeling capability is introduced into HSIC to facilitate the capture of complex relationships in HSIs [37], [38], [39]. Sun et al. [40] designed a spectral–spatial feature tokenization transformer (SSFTT) to capture spectral–spatial high-significance features. However, the excellent generalization performance of these methods presupposes sufficient training samples. Actually, obtaining high-quality agricultural data labels often requires staff to travel to the field for observation, which often comes at a high cost [41], [42]. In this regard, many methods using a small number of training samples have been proposed by scholars.

Under the condition of limited labeled samples, Huang et al. [43] proposed a spectral–spatial masked transformer (SSMTr) with a supervised and contrastive learning method. Xi et al. [44] proposed a semi-supervised graph prototype network, and Sellers et al. [45] proposed a semi-supervised method based on superpixel graph learning.

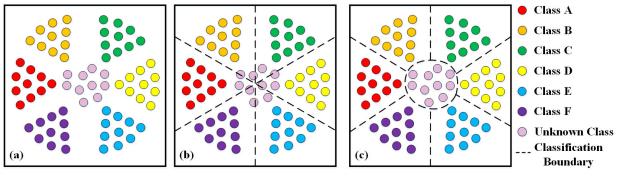


Fig. 1. Diagram of CSC and OSC. (a) Data distribution. (b) In CSC, unknown classes are randomly divided into any one of the known classes. (c) In OSC, unknown classes are distinguished.

These methods can train with a limited number of labeled samples and a majority of unlabeled samples, ensuring the generalization and robustness of the method in a small number of labeled samples, and achieving satisfactory classification results.

Because many methods with good performance have been proposed, the development of HSIC has made great progress. However, when an unknown class of data appears in the test data, existing methods will also lose their effectiveness. In other words, most existing methods are based on the closed-set assumption of ideal states: the classes of training data and test data are exactly the same [46]. However, in practical applications such as crop classification, this ideal state is almost impossible to achieve [47]. As shown in Fig. 1(a), unknown classes often exist in the tested data. The existing closed-set classification (CSC) methods cannot distinguish the features of an unknown class in the feature space. That means the data of unknown classes will be randomly divided into known classes, as shown in Fig. 1(b). In CSC, the test data are sent to the trained model, and the model relies on the softmax probability to judge what kind of test data it is. However, the CSC methods do not consider the open-set environment (OSE), so the softmax classification probability has a stubbornly closed-set property [46]. Therefore, in order to avoid the limitations of the closed-set environment (CSE) hypothesis and address the issue of the OSE, many open-set classification (OSC) methods are proposed. The diagram of an unknown class being distinguished is shown in Fig. 1(c) [46], [48]. Liu et al. [47] proposed a multitask DL method (MDL4OW), which reconstructs images to find areas that cannot be reconstructed and thus unknown class. Yue et al. [49] proposed an OSC method for spectral features implicit reconstruction, which distinguishes unknown classes by reconstructing spectral features. Xie et al. [50] proposed a feature consistency-based prototype network, which uses training data to construct a prototype set and distinguishes location classes by comparing test data with the prototype set. Sun et al. [51] proposed a spectral–spatial MLP-like network with reciprocal points learning, which combines reciprocal point learning with MLP-like networks to distinguish features of unknown class. These methods target open-set data in different scenarios and can effectively identify unknown classes in the data.

The above methods have achieved remarkable results, however, the robustness of these methods is not always maintained in agricultural scenarios. This is because different crops may have similar spectral reflectance in some bands, and the growth of the same crops may not be exactly the same. As shown in Figs. 2(b) and 3(b), the spectral reflectance curves of

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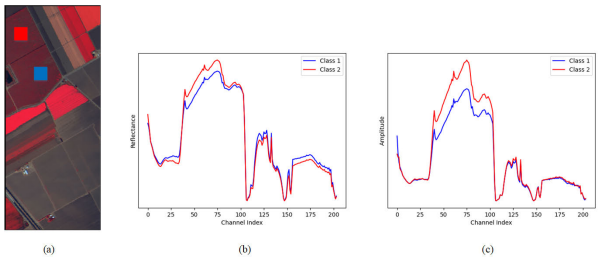


Fig. 2. FrFT renderings for different classes with similar spectral curves. (a) False-color image. (b) Spectral reflectance curve. (c) FrFT amplitude image.

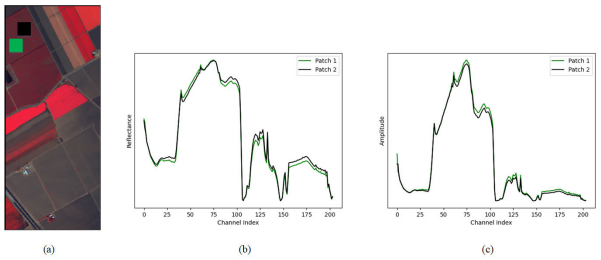


Fig. 3. FrFT renderings for same classes with different spectral curves. (a) False-color image. (b) Spectral reflectance curve. (c) FrFT amplitude Image.

different crops are similar, but the spectral reflectance curves of the same crop are not completely the same. Therefore, there are many same crops with different spectrums and many different crops with similar spectrums [52]. In order to make the distinction between spectra of different landcovers more significant, some scholars considered using frequency domain information as an auxiliary [53]. Qiao et al. [54] used curve-wave transformation and singular spectrum analysis to denoise and classify HSIs. Zhao et al. [14] proposed a fractional Gabor convolutional network for multisource remote sensing data classification. Due to its characteristic of spectral decorrelation, the fractional Fourier transform (FrFT) has an advantage in the field of HSI processing [55], [56]. In Fig. 2, the spectral reflectance curves of two patch blocks from different classes and the amplitude image obtained through FrFT are shown. Clearly, the differences between different classes of samples treated by FrFT become more obvious. In Fig. 3, the spectral reflectance curves of two patch blocks from the same class and the amplitude image obtained through FrFT are shown. After FrFT, the difference between the same samples remained the same or even became smaller. This effectively shows that FrFT can be used to obtain frequency domain information reflecting the differences of landcovers belonging to different classes.

Inspired by the above, the motivation of this article is how to make better use of frequency domain information in agricultural hyperspectral OSC. Therefore, a fractional Fourier-based frequency-spatial–spectral prototype network (FrFSSPN) is proposed to overcome the phenomenon of different crops with similar spectrum and the same crops with different spectrum in HSIs. First, the FrFT is introduced to combine the frequency-spatial–spectral information of HSIs to fully reflect the information differences between different classes. Then, the prototype learning strategy is applied to the network, and the distance between the feature vector and the prototype vector is used as the prototype loss to constrain the network, so as to enhance the integration ability of similar features and the ability to distinguish different features. Finally, the unknown class is identified by analyzing the difference between the feature vector of the test data and the prototype vector, rather than relying on softmax classification probability. The main contributions of FrFSSPN are as follows.

1) An FrFSSPN is explored for implementing open-set agricultural HSIC. The use of FrFT can effectively alleviate the phenomenon of different crops with similar spectrum and the same crops with different spectrum, and the application of multiscale and multidepth attention ensures the effective utilization of frequency-spatial–spectral information. To the best of our knowledge, this is the first time that the FrFT has been applied in the field of open-set agricultural HSIC.
2) In order to improve the ability of the network to distinguish features of different classes, a prototype learning strategy is introduced into the network. The Euclidean distance between the feature vector and the prototype vector is regarded as the prototype loss to constrain the network, which enables the network to extract more representative features rather than memory training samples.
3) In order to break the stubborn closed-set property of softmax classification probability, an open-set recognition module is proposed to effectively recognize unknown classes. The Euclidean distance between the feature vector obtained from the test data and the prototype vector is used as the evaluation criterion, and the K-means clustering is used to adaptively judge the data with large distances and label them as unknown classes.

The remaining structure of this article is as follows. Section II describes the proposed method. Section III presents the datasets and experimental setup. Section IV shows the results analysis and parameter analysis. Section V provides the conclusion.

II. Proposed Method

In this section, we describe the proposed FrFSSPN in detail, and its structure is shown in Fig. 4. The training data is divided into two parts, which are called feature data and prototype data. First, these two parts of data are processed by the frequency-spatial–spectral information integration module (FSSIIM) to integrate the frequency domain and spatial–spectral information of the data. Second, through the frequency-spatial feature extraction module and frequency-spectral feature extraction module, the final feature of the data is obtained. Among them, the feature obtained from the feature data is called the feature vector, and the feature obtained from the prototype data is called the prototype vector. Then, the prototype learning strategy is used to enhance the feature-distinguishing capability of the network. The Euclidean distance between the prototype vector and the feature vector is used as the prototype loss to the constraint network. Finally, the test data is sent to the network as feature

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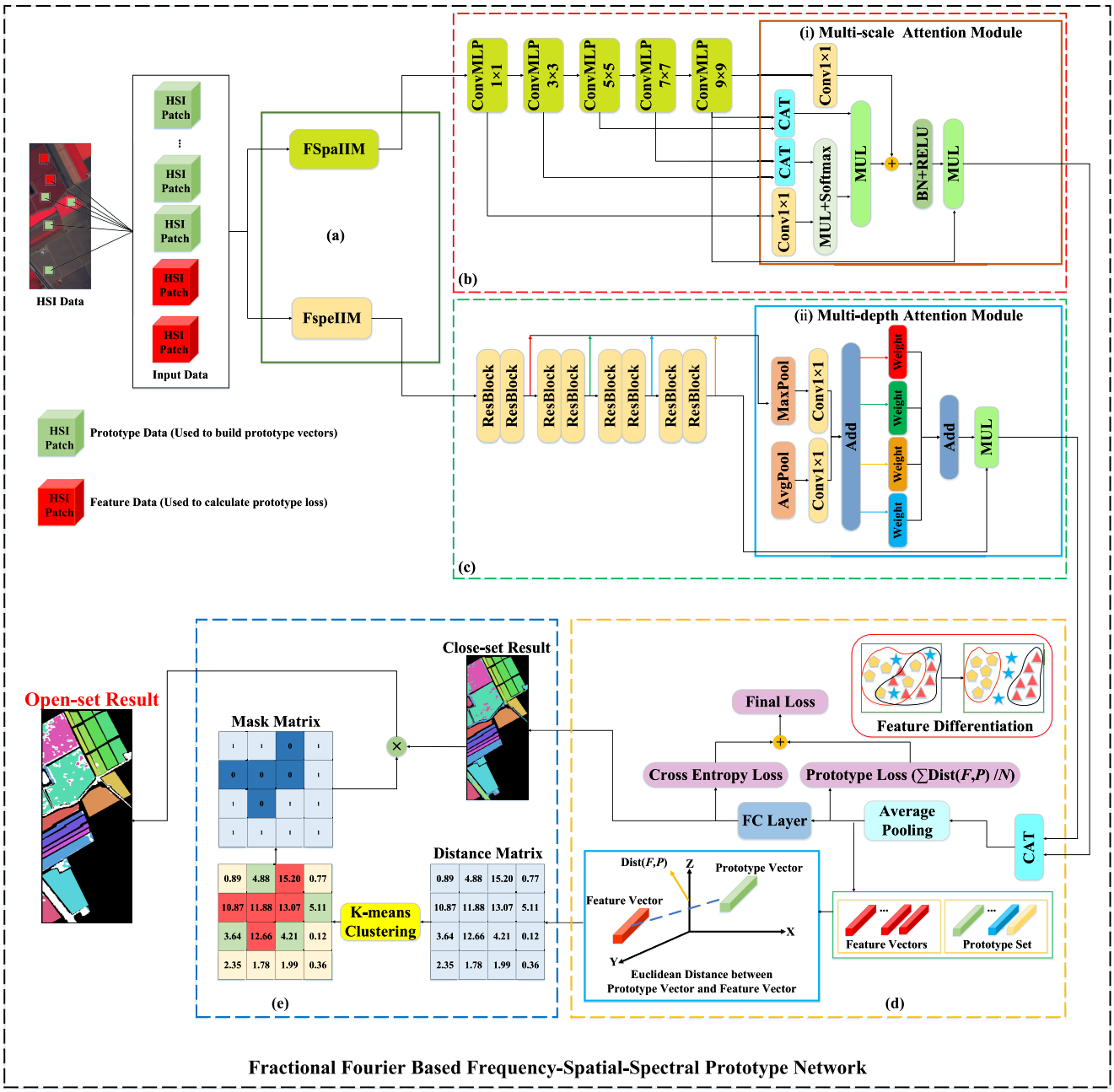


Fig. 4. Framework of the fractional Fourier based frequency-spatial–spectral prototype network. The training data is divided into two parts, one part is prototype data and the other part is characteristic data. First, these two parts of data are processed by (a) FSSIIM, (b) frequency-spatial feature extraction module, and (c) frequency-spectral feature extraction module, respectively, which can integrate the frequency domain and spatial–spectral information. Among them, the feature obtained from the feature data is called the feature vector, and the feature obtained from the prototype data is called the prototype vector. Then, (d) prototype learning strategy is applied to improve the network's ability to distinguish different features. The Euclidean distance between the prototype vector and the feature vector is used as the prototype loss to the constraint network. Finally, (e) open-set recognition module is designed for identifying unknown classes.

data, and the unknown class is identified by the open-set recognition module.

A. Frequency-Spatial–Spectral Information Integration Module

FSSIIM uses FrFT to extract the fractional frequency domain information of the image and integrate the spatial–spectral information of the image, and its structure is shown in Fig. 4(a). Aided by FrFT's ability to obtain information that reflects differences between classes. FSSIIM makes the information of images more different between classes and less different within classes. This effectively reflects the differences between different classes, and alleviates the phenomenon of different crops with similar spectrums and the same crops with different spectrums. FSSIIM includes a frequency-spatial information integration module (FSpaIIM) and a frequency-spectral information integration module (FSpeIIM).

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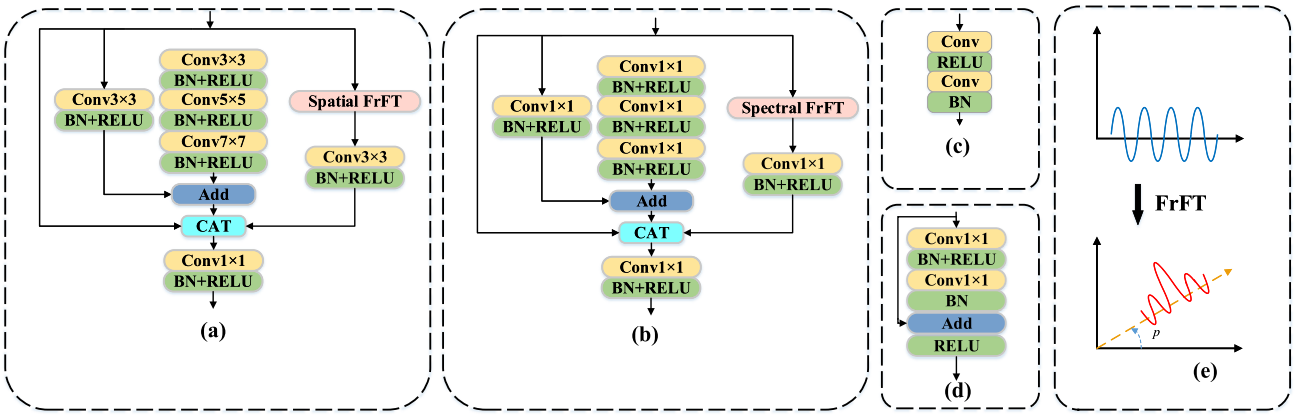


Fig. 5. Structure of related modules in FrFSSPN. (a) FSpaIIM. (b) FSpeTIIM. (c) ConvMLP model. (d) ResBlock. (e) Schematic diagram of FrFT.

1) Frequency-Spatial Information Integration Module: As shown in Fig. 5(a), in FSpaIIM, data is processed by a CNN module with three convolution layers and a CNN module with one convolution layer. The results of two CNN module calculations are added together as the shallow feature of the spatial dimension of the data. Among them, the convolution kernel size of the CNN module with three convolution layers is 3 × 3, 5 × 5, and 7 × 7, respectively, and that of the CNN module with one convolution layer is 3 × 3. In addition, in order to improve the nonlinear expression ability of the network, BatchNormal, and ReLu functions are activated after each layer of convolution. At the same time, in order to obtain the fractional frequency domain information in the spatial dimension of the data, the data is subjected to FrFT in the spatial dimension. The FrFT diagram is shown in Fig. 5(e). In the FrFT, when the fractional order changes, the signal rotates from the original signal axis to the transform domain axis in the frequency-spatial–spectral plane [55], [56], [57]. For HSIs, the FrFT of pixels is expressed as follows:

$$\mathbf{x}_p(f) = (1/d)\sum_{s=1}^{d}\mathbf{x}(s)\Omega_p(s, f) \tag{1}$$

where $\mathbf{x}$ represents a single pixel and $\Omega_p(s, f)$ is kernel function. The formula of kernel function $\Omega_p(s, f)$ is as follows:

$$\Omega_p(s, f) = \begin{cases} A_\varphi \exp\left[j\pi\left(s^2\cot\varphi - 2sf\csc\varphi + f^2\cot\varphi\right)\right] \\ \varphi \neq n\pi \\ \delta(s-f), \varphi = 2n\pi \\ \delta(s+f), \varphi = (2n\pm 1)\pi \end{cases} \tag{2}$$

where d is the channel index of the pixel, s and f are the indices of pixels, n must be an integer, and p is the order of FrFT. When $\varphi = p\pi/2$, φ is the rotation angle of the time-frequency plane. A_φ is the rotation angle of the frequency-spatial–spectral plane. The related expression is as follows:

$$A_\varphi = \frac{\exp\left[-j\pi\ \mathrm{sgn}(\sin\varphi)/4 + j\varphi/2\right]}{|\sin\varphi|^{1/2}}. \tag{3}$$

Here, when p is 0, $\mathbf{x}_p$ represents the original pixel, and when p is 1, $\mathbf{x}_p$ represents the pixel after Fourier transform. According to the order additivity of FrFT, FrFT in the two dimensions of height and width is carried out separately on the patch. The fractional Fourier frequency domain features of the spatial dimension are expressed as follows:

$$\mathrm{Feature}_{\mathrm{Spatial}} = \mathrm{BN}(\mathrm{ReLu}(\mathrm{Conv}_{3\times3}(|\ \mathcal{F}_w(\mathcal{F}_h(\mathbf{X}))\ |))) \tag{4}$$

where $\mathbf{X}$ is a patch of HSI and $\mathcal{F}_h$ and $\mathcal{F}_w$, respectively, represent the FrFT of two dimensions in space. BN represents BatchNormal.

Finally, the information of original data is combined with fractional domain information and shallow features in spatial dimension. So that the frequency-spatial information in the spatial dimension can be obtained.

2) Frequency-Spectral Information Integration Module: As shown in Fig. 5(b), the structure of FSpeIIM is basically the same as the FSpaIIM. The results of two CNN module calculations are added together as the shallow feature of the spatial dimension of the data. Among them, the convolution kernel size of all convolution layers is 1 × 1. Similarly, BatchNormal and ReLu functions are activated after each layer of convolution. At the same time, in order to obtain the fractional frequency domain information in the spectral dimension of the data, the data is subjected to FrFT in the spectral dimension. The fractional Fourier frequency domain feature of the spectral dimension is expressed as follows:

$$\mathrm{Feature}_{\mathrm{Spectral}} = \mathrm{BN}(\mathrm{ReLu}(\mathrm{Conv}_{1\times1}(|\ \mathcal{F}_c(\mathbf{X})\ |))) \tag{5}$$

where $\mathcal{F}_c$ represents the FrFT of spectral dimension.

Finally, the original information of the HSI, its spectral fractional domain information, and its spectral shallow feature are integrated together. So that the frequency-spectral information in the spectral dimension can be obtained.

B. Frequency-Spatial Feature Extraction Module

Due to its excellent ability to capture information related to spatial distribution, MLP is used to extract spatial features. The spatial feature extraction module is composed of five ConvMLP layers and a multiscale attention module (MCAM). The structure of ConvMLP is shown in Fig. 5(c). In the frequency-spatial feature extraction module, the convolution kernels of five ConvMLP are 1 × 1, 3 × 3, 5 × 5, 7 × 7,

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and 9×9, respectively. The convolution of different receptive fields can mine spatial features of different scales. Then, the features of different scales obtained by five ConvMLP modules are integrated by MCSM, and the structure of MCSM is shown in Fig. 4(i). By using MLP and multiscale attention mechanism of different scales, the influence of noise on the single scale can be reduced, and the high redundancy of HSIs can be better alleviated. At the same time, the information of different spatial scales can increase the richness of feature representation and construct nonsingle feature space, so that the model can understand the content of the image more comprehensively. Combining the diversity of feature spaces at different scales, it is beneficial to obtain more advanced classification results.

C. Frequency-Spectral Feature Extraction Module

Because of the ability to effectively learn spectral feature representation, ResNet is adopted for spectral feature extraction. The spectral feature extraction module consists of a spectral-ResNet feature extraction module and a multidepth attention mechanism module (MDAM). As shown in Fig. 4(d), the frequency-spectral feature extraction module consists of eight ResBlocks, and each ResBlock is a residual structure containing two convolution layers. The structure of ResBlock is shown in Fig. 5(d). Fig. 4(ii) shows the structure of MDAM, which takes the output sum of every two ResBlock as the attention weight and multiplies it with the feature of the final output. Using features with different depths as weights, features can be considered more comprehensively, and instead of only considering the deepest single feature space. Therefore, the network can pay more attention to the spectral information related to classification, thus effectively capturing features that are more conducive to the task [46]. HSIs have many spectral bands. The feature with different levels of abstraction can better include the relationships between different bands, thus improving the global understanding of spectral information. Low-level features can capture some basic spectral features, while high-level features can involve more complex spectral combinations and semantic information. Therefore, the features with different levels of abstraction can improve the global understanding of data in the network.

D. Prototype Learning Strategy

On the basis of ensuring the performance of CSC, the focus of the OSC task is to distinguish the characteristics of an unknown class. Based on this point, as shown in Fig. 4(d), a prototype learning strategy is introduced into the network. As mentioned earlier, in FrFSSPN, training data will be divided into prototype data and feature data. First, the prototype data is sent to the network, and the average value of feature vectors obtained from some data is called the prototype vector. This part of the data is used to calculate the cross-entropy loss to ensure the excellent feature mining ability of the network. Cross entropy loss is calculated as follows:

$$\text{Loss}_{\text{cross-entropy}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{i,j} \cdot \log\left(p_{i,j}\right) \tag{6}$$

where N is the total number of training samples and C is the total number of classes. $y_{i,j}$ is the real label that the sample i belongs to the class j. $p_{i,j}$ is the probability that the sample i belongs to the class j.

Then, the feature data is sent to the network, and the vector obtained from the feature data is called the feature vector. These vectors are calculated by Euclidean distance with the e vectors obtained from the prototype data of the same class in turn. The related formula is as follows:

$$\text{Distance} = \sqrt{\sum_{\text{k}=1}^{\text{c}}\left(V_{\text{feature[k]}} - V_{\text{prototype[k]}}\right)^2} \tag{7}$$

where c represents the number of dimensions of the prototype vector and the feature vector. $V_{\text{feature[k]}}$ represents the kth dimension of the feature vector and $V_{\text{prototype[k]}}$ represents the kth dimension of the prototype vector.

Obviously, the smaller the distance, the more similar the two vectors are, and the feature extracted by the network can better represent the same kind of data. Therefore, this distance is regarded as the sum of prototype loss and cross-entropy loss as the total loss of the whole network. The total loss expression of the network is as follows:

$$\text{Loss}_{\text{total}} = \text{Loss}_{\text{cross-entropy}} + \text{Loss}_{\text{prototype}} \tag{8}$$

where $\text{Loss}_{\text{prototype}}$ is distance.

E. Open-Set Recognition Module

To break the stubborn closed-set property of CSC methods, an open-set recognition module is proposed to identify unknown classes. First, the test data are sent to the trained model to obtain the CSC results. Second, we match the feature vectors obtained from the test data and the result labels obtained by CSC with different types of prototype vectors in the prototype set. The Euclidean distance between the same kind of feature vectors and prototype vectors is calculated, and then get a distance matrix. The relevant formula is as follows:

$$D_{H\times W} = \begin{bmatrix} \text{Dinstance}_{11} & \cdots & \text{Distance}_{1H} \\ \vdots & \ddots & \vdots \\ \text{Dinstance}_{W1} & \cdots & \text{Distance}_{HW} \end{bmatrix} \tag{9}$$

where Distance_{HW} represents the distance of the $H \times W$ pixel of the test image.

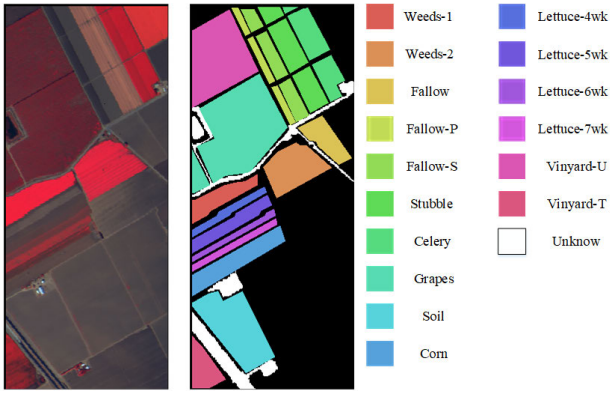
Obviously, in the distance matrix, the larger Distance_{HW}, the less similar that pixel is to the known classes. Then, the distance matrix is sent into an open-set recognition module, whose structure is shown in Fig. 4(e). In order to pick out these pixels, K-means clustering is used to divide distances into three classes. The pixels that distance is large ($\text{distance}_{\max}$) are marked as unknown classes so that the pixels that are least similar to the prototype set in the test data can be selected to achieve the purpose of OSC. Thus, a mask matrix $\text{Mask}_{H\times W}$ is obtained

$$\text{Mask}_{H\times W} = \left\{ \begin{array}{ll} 0, & \text{if Distance}_{ij} \text{ is distance}_{\max} \\ 1, & \text{else} \end{array} \right\}. \tag{10}$$

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TABLE I
LAND COVER NAME AND SAMPLE NUMBER OF THE SALINAS DATASET

Index	Class Name	Sample Number
1	Weeds-1	2009
2	Weeds-2	3726
3	Fallow	1976
4	Fallow-P	1394
5	Fallow-S	2678
6	Stubble	3959
7	Celery	3579
8	Grapes	11271
9	Soil	6203
10	Corn	3278
11	Lettuce-4wk	1068
12	Lettuce-5wk	1927
13	Lettuce-6wk	916
14	Lettuce-7wk	1070
15	Vinyard-U	7268
16	Vinyard-T	1807
17	Unknown Class	5613
	Total	59742



(a) (b)

Fig. 6. (a) False-color image and (b) ground-truth of the Salinas dataset.

Finally, the mask matrix is multiplied with the closed-set result, and the OSC result is obtained. The relevant formula is as follows:

$$\text{Result}_{\text{Open-set}} = \text{Mask}_{H\times W} \times \text{Result}_{\text{Closed-set}}. \quad (11)$$

III. DATASETS AND EXPERIMENTAL SETUP

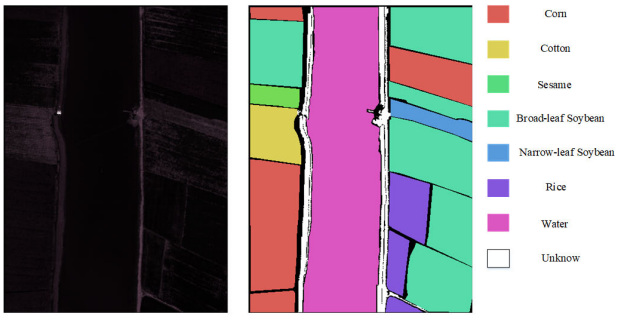
A. Datasets

1) Salinas: This is an image obtained by an AVIRIS sensor in the Senaris Valley, CA, USA. It has 224 bands with a band range of 0.4–2.5 μm, a spatial resolution of 3.7 m, and an image size of 512 × 217, including 16 ground objects. The pool in the image and some areas covered by artificial materials are marked as unknown class [47]. The challenge of this image is that there are similar crops in different growth cycles and many different crops with similar spectrums. Details of the Salinas dataset are shown in Table I. The false-color image and ground truth of Salinas are shown in Fig. 6.

2) WHU-Hi-LongKou (LK): The WHU-Hi-LongKou dataset [58], [59] was acquired in Longkou Town, Hubei province, China, with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) UAV platform. It has 270 bands, the spatial resolution is 0.463 m, and the influence size is 550 × 400, including nine ground objects. Among them, weeds, roads, and houses are marked as unknown class. The challenge of this image is that there are crops with different leaf sizes and the growth of the same crop is different. Details of the LK dataset are shown in Table II. The false-color image and ground truth of LK are shown in Fig. 7.

TABLE II
LAND COVER NAME AND SAMPLE NUMBER OF THE LK DATASET

Index	Class Name	Sample Number
1	Corn	34511
2	Cotton	8374
3	Sesame	3031
4	Broad-leaf Soybean	63212
5	Narrow-leaf Soybean	4151
6	Rice	11854
7	Water	67056
8	Unknow	12353
	Total	204542



(a) (b)

Fig. 7. (a) False-color image and (b) ground-truth of the LK dataset.

TABLE III
LAND COVER NAME AND SAMPLE NUMBER OF THE TEA DATASET

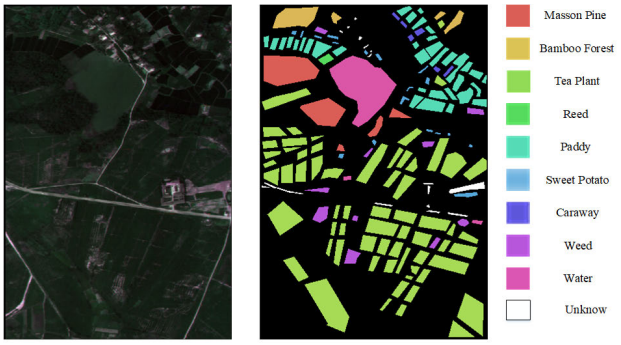
Index	Class Name	Sample Number
1	Masson Pine	5806
2	Bamboo Forest	2318
3	Tea Plant	28428
4	Reed	214
5	Paddy	6809
6	Sweet Potato	817
7	Caraway	429
8	Weed	1861
9	Water	6141
10	Unknow	911
	Total	53734

3) Tea Planting Base (Tea): The image [60] was taken at the tea planting base in Fanglu Village, Changzhou City, Jiangsu Province, and passed through the aerial push-broom hyperspectral imager push-broom hyperspectral imager (PHI). The data contains 348 × 512 pixels with a spatial resolution of 2.25 m and 80 spectral bands. It contains ten types of ground objects, among which the road and building classes are set to unknown classes. The challenges of this image are the unbalanced distribution of classes and the small proportion of unknown classes. Details of the Tea dataset are shown in Table III. The false-color image and ground truth of Tea are shown in Fig. 8.

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TABLE IV
EXPERIMENTAL RESULTS ON THE SALINAS DATASET (%)

Class	3DCNN	SSRN	DBDA	CDC_MDAA	SSMTr	SSFTT	MDL4OW	OURS
Weeds-1	99.25±0.58	**100±0**	75.17±28.07	97.04±4.08	99.35±0.94	99.95±0.06	98.38±0.02	99.93±0.11
Weeds-2	99.34±0.27	99.59±0.39	99.59±0.80	**99.89±0.14**	96.12±5.54	99.61±0.33	97.69±0.08	99.12±0.44
Fallow	98.70±0.66	99.88±0.24	80.48±21.43	98.08±2.62	98.50±1.34	**100±0**	97.67±0.04	99.56±0.54
Fallow-P	98.21±1.09	**99.67±0.26**	97.93±1.40	99.27±0.33	99.51±0.29	99.35±0.56	98.85±0.11	99.52±0.42
Fallow-S	94.98±1.12	**99.17±0.38**	95.94±4.44	97.63±1.31	98.07±0.87	96.76±0.55	91.75±0.21	96.85±1.24
Stubble	98.12±1.73	99.75±0.46	99.81±0.23	99.06±0.98	99.81±0.22	**99.95±0.09**	93.67±0.06	99.28±0.76
Celery	98.79±0.71	99.61±0.51	78.12±39.10	99.14±0.69	98.55±0.36	**99.97±0.13**	93.34±0.13	99.39±0.46
Grapes	75.85±2.02	60.31±25.60	55.62±39.64	60.38±34.83	83.03±6.93	85.23±3.41	59.24±6.81	**90.54±1.96**
Soil	98.38±0.82	99.74±0.19	96.75±5.31	**99.98±0.03**	99.42±0.44	99.95±0.06	98.87±0.17	99.70±0.42
Corn	92.53±3.17	96.01±1.73	78.54±19.33	86.11±3.88	92.78±2.96	**97.15±1.25**	92.32±0.18	91.36±7.01
Lettuce-4wk	98.89±0.57	98.24±2.07	81.82±35.53	95.09±1.95	94.01±2.57	**99.94±0.07**	99.69±0.08	98.35±2.11
Lettuce-5wk	98.56±0.85	88.77±22.46	96.80±2.76	99.95±0.06	**100±0**	99.93±0.07	99.97±0.02	99.44±0.84
Lettuce-6wk	96.55±1.95	99.91±0.13	92.21±9.27	99.30±0.63	**99.93±0.09**	99.78±0.29	98.83±0.06	99.33±1.13
Lettuce-7wk	94.51±2.45	99.08±1.05	97.10±3.88	98.67±0.29	99.53±0.42	97.81±2.66	97.32±0.16	**99.67±0.69**
Vinyard-U	71.38±5.60	75.85±37.41	56.31±26.98	48.62±40.92	61.76±11.61	83.58±4.53	87.98±1.27	**92.08±3.79**
Vinyard-T	96.74±0.86	84.19±28.82	94.11±9.76	98.39±0.49	94.39±1.98	97.86±0.96	97.78±0.12	**98.49±1.22**
Unknown class	7.79±2.12	68.92±6.50	40.89±8.24	47.79±12.23	17.97±2.48	59.85±11.02	74.84±0.98	**88.87±3.23**
OA	81.73±0.82	85.40±4.11	75.41±7.32	80.03±2.05	83.18±1.19	90.97±1.21	86.94±0.31	**95.18±0.52**
AA	89.33±0.59	92.28±4.43	83.36±6.59	89.67±1.09	90.16±0.74	95.09±0.80	93.03±0.11	**96.26±1.81**
KC	80.02±0.90	83.95±4.61	73.11±7.79	78.09±2.21	81.53±1.32	90.08±1.33	85.69±0.33	**94.70±0.57**



(a) (b)

Fig. 8. (a) False-color image and (b) ground-truth of the Tea dataset.

B. Experimental Setup

The overall accuracy (OA), average accuracy (AA), and Kappa coefficient (KC) are utilized to analyze the superiority of the proposed method. Six classical methods are employed for comparison, including CSC methods 3DCNN [31], SSRN [32], DBDA [35], CDC-MDAA [36], SSFTT [40], SSMTr [43], and open-set method MDL4OW [47]. The universal training parameters are set the same for all methods, and each method's specific parameters are configured according to the original literature. All methods employ 40 pixels per class for training. Specifically, in FrFSSPN, 35 pixels per class are used as feature data and five pixels per class serve as prototype data. Furthermore, all experiments are conducted ten times, and the experimental results are presented in the form of "mean ± standard deviation." For closed-set methods, a softmax probability less than 0.5 is considered an unknown class [47], [49], [50], [51]. In addition, all experiments were conducted on the same computer with the following specifications: CPU-Intel Xeon Silver 4210, RAM-32GB, GPU-GTX 3090 with 24 GB VRAM.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

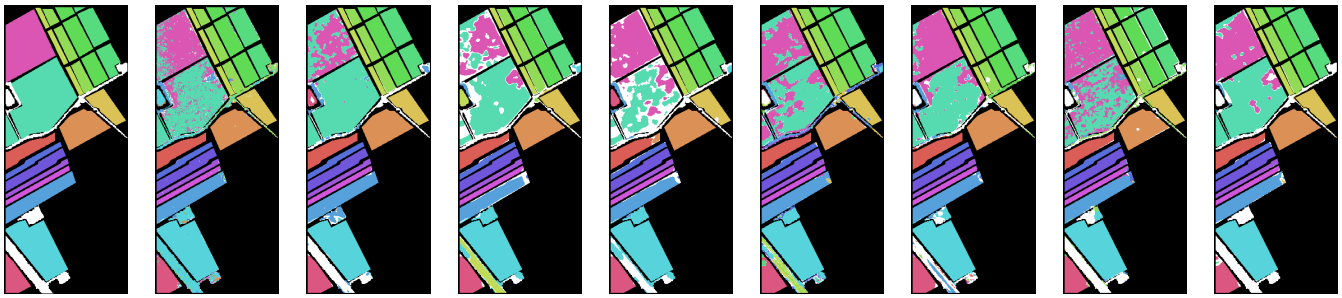
In this section, the experimental results of all methods on the three datasets are detailed in the form of figures and tables. At the same time, the ablation experiments of the FrFTFPN and the influence analysis of relevant parameters on the experimental results are presented in detail.

A. Result Analysis

1) Experimental Results on the Salinas Dataset: In the first experiment, all the methods were applied to the Salinas dataset. Table IV shows the OA, AA, KC, and the classification accuracy of each class of all methods. Under the same experimental conditions, the OA, AA, and KC of FrFSSPN are the highest, with an average increase of 5%. Among them, OA is 95.18%, AA is 96.26% and KC is 94.70%. In addition, the standard deviation of FrFSSPN experimental results is also relatively small, which indicates its stability. At the same time, for unknown classes, FrFSSPN obtained a classification accuracy of 90.79%, which is greatly improved compared with MDL4OW. In addition, FrFSSPN also achieved good classification results for the same crops in different growth cycles (Lettuce-4WK, Lettuce-5WK, Lettuce-6WK, and Lettuce-7WK) and different crops with similar spectrum (Grapes, Vinyard-U, and Vinyard-T). In the FrFSSPN results, these class accuracy has not all reached the highest, this is because the pixels at the edge have serious spectral obfuscation characteristics, so they were judged as unknown classes. However, because HSIs have a relatively low spatial resolution, in practice, these small errors do not affect the analysis of the overall situation of the crop. Fig. 9 shows the classification result graphs of all methods on the Salinas dataset, and obviously the result graphs of FrFSSPN show its good classification performance.

2) Experimental Results on the LK Dataset: In the second experiment, all the methods were applied to the LK dataset. Table V shows the OA, AA, KC, the classification accuracy of each class, and the running time of all methods. Under the same experimental conditions, the OA, AA, and KC of FrFSSPN are the highest, with an average increase of 2.5%. Among them, OA is 95.52%, AA is 95.15% and

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 9. Classification maps on Salinas dataset. (a) Ground truth. (b) 3DCNN. (c) SSRN. (d) DBDA. (e) CDC-MDAA. (f) SSMTr. (g) SSFTT. (h) MDL4OW. (i) OURS.

TABLE V
EXPERIMENTAL RESULTS ON THE LK DATASET (%)

Class	3DCNN	SSRN	DBDA	CDC-MDAA	SSMTr	SSFTT	MDL4OW	OURS
Corn	86.67±3.23	98.97±0.57	99.00±0.37	98.48±0.05	98.81±0.52	**99.27±0.14**	98.50±0.14	97.64±0.58
Cotton	74.75±5.47	97.77±1.01	**99.32±0.58**	96.07±1.98	90.98±3.32	98.66±0.61	83.00±0.95	94.31±0.86
Sesame	95.92±0.65	99.10±0.56	99.14±0.47	99.41±0.32	91.85±14.15	**99.93±0.07**	95.07±0.17	94.60±2.53
Broad-leaf Soybean	52.71±4.44	90.08±4.83	**93.29±1.75**	89.66±4.77	88.07±3.18	91.79±0.66	93.07±0.47	90.95±2.35
Narrow-leaf Soybean	82.62±2.74	**97.07±2.69**	96.97±1.19	94.40±6.51	91.71±7.41	95.52±1.81	94.90±0.24	94.91±1.93
Rice	81.17±3.09	**99.34±0.07**	99.22±0.23	99.29±0.11	95.92±5.17	97.88±0.93	97.14±0.25	96.43±1.00
Water	97.17±1.15	99.70±0.09	99.65±0.10	**99.93±0.05**	99.41±0.03	99.03±0.35	99.37±0.06	99.24±0.38
Unknown class	18.09±0.50	54.97±3.84	33.30±9.36	61.26±8.86	6.23±1.91	13.93±3.10	46.26±0.55	**93.13±2.54**
OA	74.74±1.54	93.74±1.34	93.47±1.14	93.79±0.97	89.36±1.64	91.56±0.33	93.12±0.14	**95.52±0.64**
AA	73.64±0.66	92.13±0.49	89.99±1.45	92.27±0.96	82.87±3.91	87.01±0.56	88.42±0.13	**95.15±0.33**
KC	67.47±1.82	91.85±1.70	91.15±1.48	91.92±1.22	86.14±2.11	88.95±0.43	90.99±0.17	**94.16±0.83**

KC is 94.16%. For unknown classes, FrFSSPN obtained a classification accuracy of 93.13%, which is the highest among all methods. In response to the challenge of the LK dataset, FrFSSPN can effectively identify unknown classes and obtain good classification results in other classes. Fig. 10 shows the classification result graphs of all methods on the LK dataset. In the LK dataset, some crops have not yet grown or have not obvious growth. By comparing the false-color map and the resulting map, it can be seen that FrFSSPN classifies those crops into unknown classes, which is of great significance for crop analysis.

3) Experimental Results on the Tea Dataset: In the third experiment, all the methods were applied to the Tea dataset. Table VI shows the OA, AA, KC, the classification accuracy of each class, and the running time of all methods. Under the same experimental conditions, the OA, AA, and KC of FrFSSPN are the highest, with an average increase of 2%. Among them, OA is 93.49%, AA is 94.87% and KC is 90.44%. For unknown classes, FrFSSPN obtained a classification accuracy of 87.44%, which is the highest among all methods. In response to the challenge of the Tea dataset, FrFSSPN achieved good classification results in other classes on the premise of achieving good classification accuracy of the location class and did not obtain poor accuracy in a certain class due to the unbalanced distribution of the classes. This shows that FrFSSPN can ensure reliable analysis ability in the case of uneven distribution of crop classes. Fig. 11 shows the classification result graphs of all methods on the Tea dataset. It is noteworthy that for unknown classes, all closed-set methods achieve the worst accuracy on the Tea dataset. This is because the Tea dataset is more complex than the other two datasets. Specifically, on the Tea dataset, the unknown class has fewer pixels and more dispersed spatial locations. In addition, on the Tea dataset, the proportion of pixels in each class is more uneven.

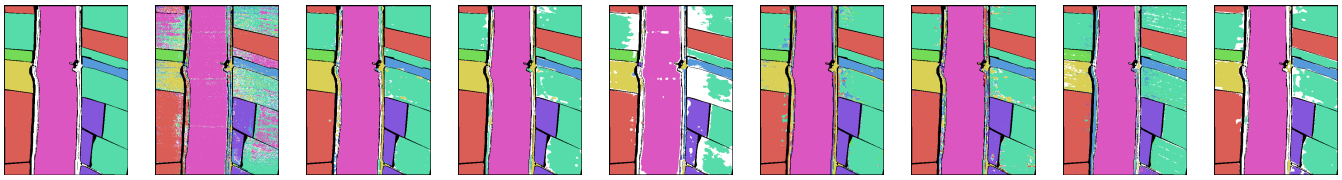
4) Running Time: The running time of all methods on the three datasets is shown in Table VII. Of all the methods, 3DCNN has the shortest running time because it has the simplest network structure. MDL4OW runs the longest because its open-set recognition part requires mathematical distribution calculations. FrFTFPN maintains a moderate uptime at optimum performance.

Since the CSC method does not use the threshold method to distinguish unknown classes, it can be seen from the experimental results that some CSC methods can distinguish unknown classes. This means that some CSC methods have certain OSC abilities and can distinguish features to a certain extent. The experimental results on the three datasets show that FrFSSPN achieves more advanced classification performance, especially for the classification of unknown classes.

B. Ablation Experiment

In order to verify the rationality of FrFSSPN, we conducted ablation experiments on three main modules of FrFSSPN. The experimental results are shown in Tables VIII–X. Under the same conditions, FSSIIM, prototype learning strategy, and open-set recognition module are, respectively, used to verify the validity of FrFSSPN. In the tables, modules that are used are marked as ✓, and modules that are not used are marked as ×. The other experimental Settings remained unchanged. Especially, when the open-set recognition module is not used,

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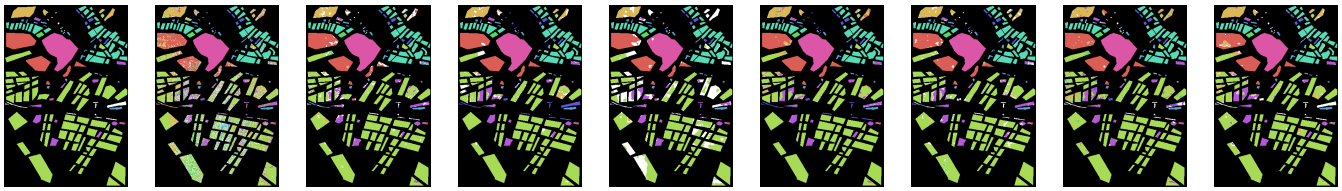
(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 10. Classification maps on LK dataset. (a) Ground truth. (b) 3DCNN. (c) SSRN. (d) DBDA. (e) CDC-MDAA. (f) SSMTr. (g) SSFTT. (h) MDL4OW. (i) OURS.

TABLE VI

EXPERIMENTAL RESULTS ON THE TEA DATASET (%)

Class	3DCNN	DBDA	SSRN	CDC-MADD	SSMTr	SSFTT	MDL4OW	OURS
Masson Pine	76.67±5.81	**97.58±1.39**	95.77±1.80	94.95±2.80	94.59±2.05	96.21±1.75	95.49±0.68	95.15±4.39
Bamboo Forest	57.09±7.79	71.98±7.38	86.75±5.51	62.62±4.12	82.86±6.03	86.25±2.81	86.43±1.02	**96.93±1.12**
Tea Plant	60.16±4.31	90.46±1.37	90.55±2.49	88.71±2.73	**92.62±1.83**	92.55±1.16	90.05±0.71	90.33±2.28
Reed	96.17±3.62	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	99.77±0.41	**100±0**
Paddy	96.93±0.38	98.88±0.90	99.43±0.11	98.91±0.39	99.22±0.19	**99.58±0.26**	99.15±0.07	98.75±0.46
Sweet Potato	90.09±2.65	95.18±0.99	97.14±1.66	93.95±1.29	98.61±1.06	**98.61±0.90**	79.47±0.22	94.05±3.22
Caraway	97.95±1.45	99.81±0.27	99.49±0.52	99.25±0.74	99.77±0.15	**100±0**	99.59±0.10	95.15±1.28
Weed	56.85±4.65	87.86±5.46	92.22±3.22	75.01±13.24	**94.38±2.93**	87.55±1.59	92.65±0.50	92.39±8.71
Water	98.27±0.66	98.82±0.01	98.98±0.45	98.98±0.25	**99.49±0.52**	99.37±0.51	98.78±0.01	98.50±0.41
Unknown class	5.05±1.97	15.17±24.03	24.24±18.46	9.73±4.44	2.39±1.48	5.67±2.25	68.80±2.72	**87.44±7.38**
OA	70.68±2.44	91.27±0.82	92.18±1.03	89.12±2.02	92.74±0.66	92.89±0.64	92.32±0.44	**93.49±0.37**
AA	73.52±1.76	85.57±2.20	88.46±2.51	82.21±1.60	86.39±0.88	86.59±0.22	91.02±0.32	**94.87±1.30**
KC	62.20±2.66	87.57±1.11	88.86±1.38	84.64±2.69	89.55±0.84	89.77±0.88	89.04±0.61	**90.44±0.88**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 11. Classification maps on Tea dataset. (a) Ground truth. (b) 3DCNN. (c) SSRN. (d) DBDA. (e) CDC-MDAA. (f) SSMTr. (g) SSFTT. (h) MDL4OW. (i) OURS.

TABLE VII

RUNNING TIME OF ALL METHODS ON THREE DATASETS (SECONDS)

Class	3DCNN	SSRN	DBDA	CDC_MDAA	SSMTr	SSFTT	MDL4OW	OURS
Salinas	35	63	58	70	87	69	93	84
LK	47	40	62	78	94	83	108	93
Tea	33	59	54	67	81	66	85	77

TABLE VIII

ABLATION EXPERIMENT RESULT ON THE SALINAS DATASET (%)

Data	FSSIIM	Prototype Learning	Open-set Recognition Module	OA	AA	KC	Unknown class
Salinas	✓	✓	✓	**95.18**	**96.26**	**94.70**	**88.87**
	✓	×	✓	90.69	95.45	93.50	87.80
	✓	×	×	90.97	94.81	90.09	78.55
	✓	✓	×	90.09	94.84	89.14	82.74
	×	✓	✓	86.77	92.81	85.52	75.15
	×	×	✓	84.16	91.37	82.69	61.69
	×	✓	×	86.32	91.45	84.89	55.67
	×	×	×	86.54	92.01	85.24	54.91

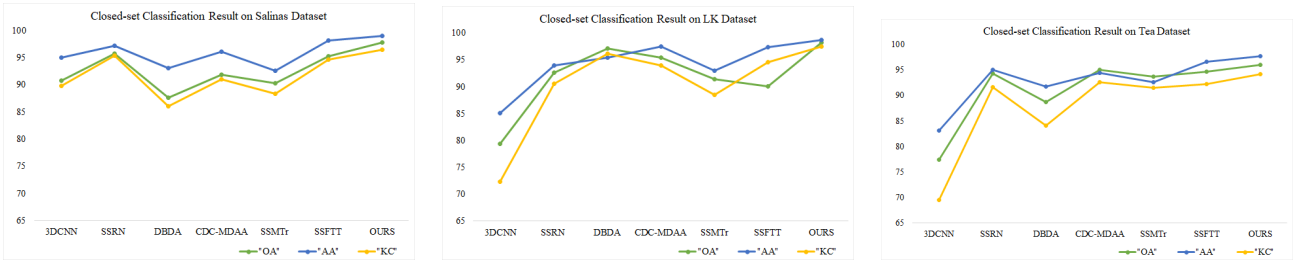
the samples with softmax classification probabilities less than 0.5 are considered unknown classes.

As can be seen from the tables, FrFSSPN has the best classification performance. Under the premise of keeping other conditions equal, as the most important part, when FSSIIM is not used, the classification performance of the method decreases obviously, especially the classification accuracy of unknown class. This shows that the frequency domain information obtained by FrFT can effectively improve the quality of the features extracted by the network. In addition, when only FSSIIM is used, the classification effect is also significantly better than that without FSSIIM. This also shows the rationality of FrFT to improve the performance of agricultural OSC. Under the premise of keeping other conditions equal, when the prototype learning strategy is used, the classification performance of the method improves. Especially under the premise of using FSSIIM, the positive effect of the prototype learning strategy is more obvious. This not only shows the effectiveness of the prototype learning strategy but also shows that the differences between classes and the similarities within the same category can be well discovered by using the frequency domain information. In addition, the open-set recognition module plays an obvious role as the main module for identifying unknown classes. It can be seen from the tables that when the open-set recognition module is used, the classification accuracy of an unknown class is the highest and significantly improved under the same conditions.

C. CSC Experiment

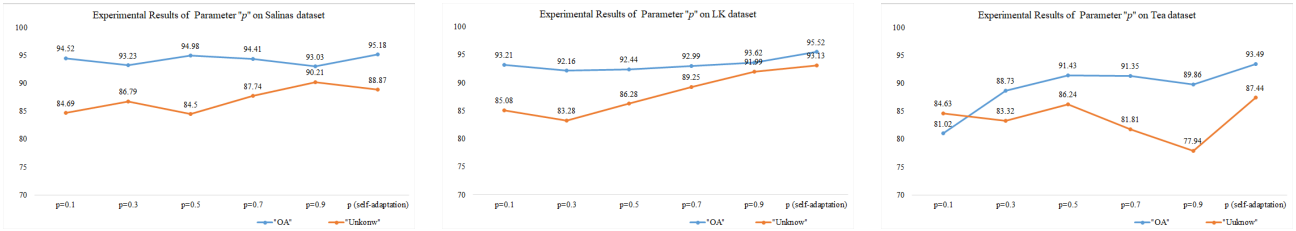
Being able to effectively identify unknown classes is the focus of OSC tasks. However, the task is meaningless if the

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(a) (b) (c)

Fig. 12. CSC result. (a) Salinas. (b) LK. (c) Tea.



(a) (b) (c)

Fig. 13. Experimental results of parameter p on three datasets. (a) Salinas. (b) LK. (c) Tea.

TABLE IX

ABLATION EXPERIMENT RESULT ON THE LK DATASET (%)

Data	FSSIIM	Prototype Learning	Open-set Recognition Module	OA	AA	KC	Unknown class
LK	✓	✓	✓	**95.52**	**95.15**	**94.16**	**93.13**
	✓	×	✓	91.38	93.57	92.91	89.59
	✓	×	×	91.66	93.47	92.39	80.32
	✓	✓	×	90.87	92.91	91.44	81.57
	×	✓	✓	87.33	90.24	89.11	78.38
	×	×	✓	85.49	89.17	88.57	65.35
	×	✓	×	87.23	89.21	88.71	53.54
	×	×	×	88.47	89.04	86.16	51.36

TABLE X

ABLATION EXPERIMENT RESULT ON THE TEA DATASET (%)

Data	FSSIIM	Prototype Learning	Open-set Recognition Module	OA	AA	KC	Unknown Class
Tea	✓	✓	✓	**93.49**	**94.87**	**90.44**	**87.44**
	✓	×	✓	89.02	90.75	85.02	70.51
	✓	×	×	89.34	91.18	85.48	66.16
	✓	✓	×	90.86	91.84	87.42	76.51
	×	✓	✓	87.06	88.73	82.46	55.63
	×	×	✓	87.60	86.36	83.05	49.78
	×	✓	×	85.63	85.42	80.39	38.55
	×	×	×	85.44	85.14	80.28	36.26

classification performance of CSC is thus sacrificed. Therefore, FrFSSPN and other CSC methods were tested with CSC experiments on three datasets. Here, we remove the open-set recognition module of FrFSSPN and keep the rest for CSC. Other CSC methods were used to perform CSC experiments normally. Fig. 12 shows the CSC results of all methods on three datasets. It can be seen from the figure that OA, AA, and KC of FrFSSPN are superior to other methods under the same experimental conditions. Other methods show inconsistent performance on different datasets. In the case of DBDA, on the Salinas dataset, its performance is lower than that of SSRN. However, on the LK dataset, it performs better than SSRN. This also fully verifies that FFRN has good robustness and generalization. This fully shows that FrFSSPN is not only concerned with the classification effect of unknown classes but also obtains a good open-set effect on the premise of ensuring the classification performance of closed-set.

D. Parameter Analysis

1) Order of FrFT p: In the FrFT, the original frequency domain characteristics of images change with the change of order p [55]. For FrFSSPN, in Sectral-DFTIIM, 1-D FrFT is performed in the spectral dimension, and in FSpaIIM, 2-D FrFT is performed in the spatial dimension. For FrFT of different dimensions, the choice of p must be different. Therefore, in FrFT, we set p as an adaptive parameter constrained by training loss. Several experiments are performed on three datasets to demonstrate the effect of p on method performance. Fig. 13 shows the relevant experimental results, OA, and the accuracy of an unknown class under different p is shown in the form of line charts. When we set the p of spectral dimension and spatial dimension to the same, different p corresponds to different classification effects. It is worth noting that in some cases, the accuracy of the unknown class is the highest, but the OA is not the highest at this time. This suggests that the high accuracy of the unknown class classification is achieved at the expense of the accuracy of other known classes at this time. Therefore, it is desirable to set p differently in spectral dimension and spatial dimension. When p is adaptive, OA is the highest, and the accuracy of the unknown class is also maintained at a high level. This fully demonstrates the validity and rationality of adaptive p.

2) Number of Clustering Centers for K-Means: In the open-set recognition module, the Euclidean distance between the test data feature vector and the prototype vector is used as the criterion to judge whether it is an unknown class. K-means clustering is used to generate mask matrices to obtain open-set results. As a rule of thumb, in OSC, the centroid number of K-means can be set to 2, that is, the test data is divided into unknown and known classes. However, due to the spectral redundancy of HSIs, the characteristics of similar features are not the same, or even quite different. These results in some

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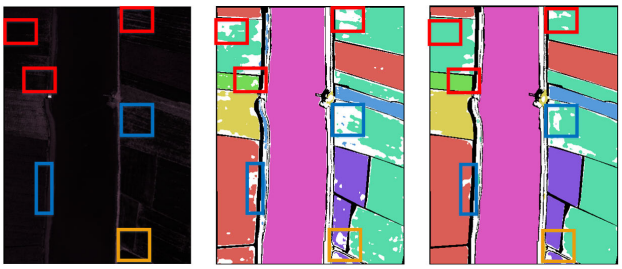


Fig. 14. Experimental results of different centroid numbers of K-means on the LK dataset. (a) False-color image. (b) Centroid number is 2. (c) Centroid number is 3.

samples in the test data being classified as unknown, when in fact they belong to known classes. To solve this phenomenon, we set the centroid number of K-means clustering to 3. The part of the samples with the largest distance from the prototype vector is considered to be an unknown class, and the others are known classes. This can effectively classify the known class with large feature differences correctly. The relevant situation is shown in Fig. 14. When we look at the false-color images with our eyes, there are obvious differences between some of the same features. When the centroid number is set to 2, these very different pixels are considered to be unknown classes. However, when the centroid number is set to 3, the situation is significantly improved. This also fully illustrates the advantages of FrFSSPN.

V. Conclusion

In this article, a FrFSSPN is proposed. so that the phenomenon of different crops with similar spectra and the same crops with different spectrum can be effectively alleviated. Furthermore, the application of a prototype learning strategy can effectively improve the network's ability to distinguish features, so that the network can better extract features instead of memorizing training samples. In addition, the open-set recognition module breaks the closed-set restriction and distinguishes the unknown class by distinguishing the difference between the feature vector and the prototype vector. Experiments on three agricultural HSI datasets validate FrFSSPN's advanced classification performance, especially for unknown classes.

A large number of experiments show that combining the frequency domain information of HSI with its spatial–spectral information can effectively improve classification performance. Frequency domain information and spatial–spectral information can complement each other's shortcomings, and more effectively search for features of different classes and more authoritative integration of features within the same class, especially in agricultural scenarios. At the same time, the phenomenon that different crops with similar spectra and the same crops with different spectrum can be effectively alleviated in the frequency domain. In future work, we will focus on how to better combine frequency-domain information with spatial–spectral information and apply it to other fields. In addition, the speed of the operation of the method is also the direction of future work.

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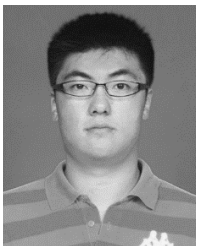
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